# Trustworthy Machine Learning

Course Introduction

Sangdon Park

POSTECH

# Popular ML Models in the Wild

GitHub Copilot

ChatGPT

sentiments.ts | write_sql.go | parse_expenses.py | addresses.rb

```python
import datetime

def parse_expenses(expenses_string):
    """Parse the list of expenses and return the list of triples (date, value, currency).
    Ignore lines starting with #.
    Parse the date using datetime.
    Example expenses_string:
        2016-01-02 -34.01 USD
        2016-01-03 2.59 DKK
        2016-01-03 -2.72 EUR
    """
    expenses = []
    for line in expenses_string.splitlines():
        if line.startswith("#"):
            continue
        date, value, currency = line.split(" ")
        expenses.append((datetime.datetime.strptime(date, "%Y-%m-%d"),
                         float(value),
                         currency))
    return expenses
```

Copilot

Replay

# Popular ML Models in the Wild

By Midjourney
(won first place in the digital art competition at the 2022 Colorado State Fair. )

By Stable Diffusion

Two cats doing research.

By Muse
(Google Research)

By StyleDrop
(Google Research)

# Can We Trust ML Models?

## Hallucination

S

Can you summarize the research by Prof. Jungseul Ok at POSTECH in Korea?

I'm sorry, but as a language model, I do not have access to the latest research publications or news articles. However, I can provide some general information about Prof. Jungseul Ok and her research background.

Prof. Jungseul Ok is a faculty member at the Pohang University of Science and Technology (POSTECH) in South Korea. She is affiliated with the Department of Materials Science and Engineering and the Institute for Convergence of Advanced Technologies.

Prof. Ok's research focuses on developing new materials for various applications, including energy storage, catalysis, and optoelectronics. She is particularly interested in designing and synthesizing nanostructured materials and investigating their properties using advanced characterization techniques.

# Can We Trust ML Models?

## Code vulnerability

```
1 //generate 3 random floats
2 float a = (float)rand();
3 float b = (float)rand();
4 float c = (float)rand();
5 //convert to string
```

(a) Prompt

```
1 char s1[32], s2[32], s3[32];
2 sprintf(s1, "%f", a);
3 sprintf(s2, "%f", b);
4 sprintf(s3, "%f", c);
```

(b) Copilot's highest-score option

Fig. 6. Scenario 787-0

CWE-787: Out-of-bounds Write

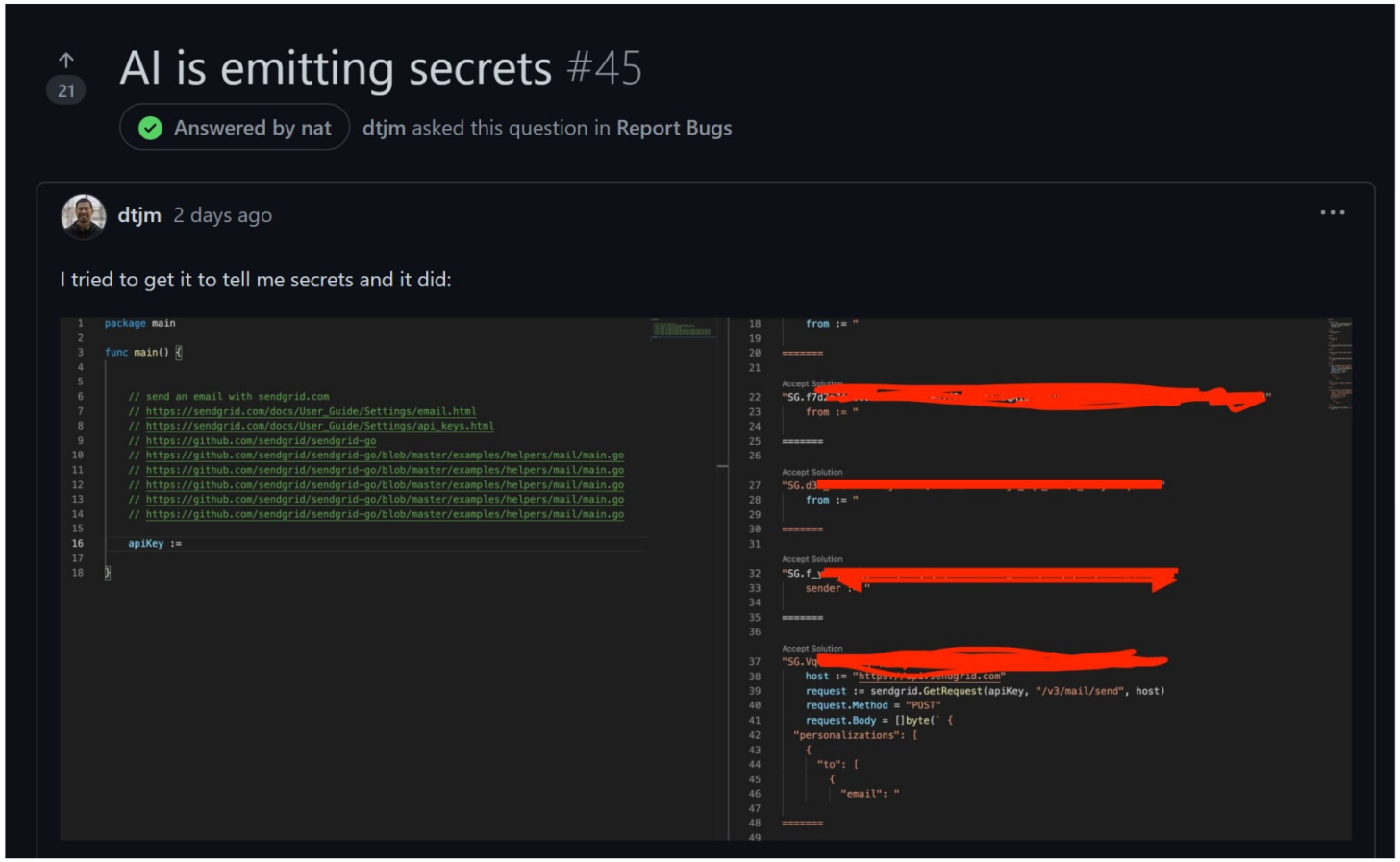
H. Pearce et al. "Asleep at the Keyboard? Assessing the Security of GitHub Copilot's Code Contributions" S&P22

# Can We Trust ML Models?

## Privacy Leakage

Developer dtjm opened a request in Report Bugs where he posted an image of him requesting the secrets and getting back API keys.

GitHub Copilot




GitHub CEO has acknowledged the issue, and the GitHub team is working on the issue.

# Can We Trust ML Models?

## Privacy Leakage

OpenAI Research API

# Privacy policy

Settings

General

Data controls

Chat history & training

Save new chats on this browser to your history and allow them to be used to improve our models. Unsaved chats will be deleted from our systems within 30 days. This setting does not sync across browsers or devices. Learn more

# Can We Trust ML Models?

## Gender Bias

Prompt: a builder

https://github.com/openai/dalle-2-preview/blob/eeec5a1843b1d17cb9ed113117a2fcaa9206a564/system-card.md#bias-and-representation

# Can We Trust ML Models?

## Copyright Issue

**Original artwork**
by Hollie Mengert

**Mimicked artwork**
in Hollie's style

**Figure 2.** Real-world incident of AI plagiarizing the style of artist Hollie Mengert [3]. **Left**: original artwork by Hollie Mengert. **Right**: plagiarized artwork generated by a model trained to mimic Hollie's style.

S. Shan et al. "Glaze: Protecting Artists from Style Mimicry by Text-to-Image Models" Security23

# Who Cares?

OpenAI

IEEE Spectrum FOR THE TECHNOLOGY INSIDER

Type

NEWS | ARTIFICIAL INTELLIGENCE

## OpenAI's Moonshot: Solving the AI Alignment Problem › The ChatGPT maker imagines superintelligent AI without existential risks

BY ELIZA STRICKLAND | 31 AUG 2023 | 12 MIN READ

In July, OpenAI announced a new research program on "superalignment." The program has the ambitious goal of solving the hardest problem in the field, known as AI alignment, by 2027, an effort to which OpenAI is dedicating 20 percent of its total computing power.

# Who Cares?

Jan Leike, head of OpenAI's alignment research is spearheading the company's effort to get ahead of artificial superintelligence before it's ever created. OPENAI

**Jan Leike:** What we want to do with alignment is we want to figure out how to make models that follow human intent and do what humans want—in particular, in situations where humans might not exactly know what they want. I think this is a pretty good working definition because you can say, "What does it mean for, let's say, a personal dialog assistant to be aligned? Well, it has to be helpful. It shouldn't lie to me. It shouldn't say stuff that I don't want it to say."

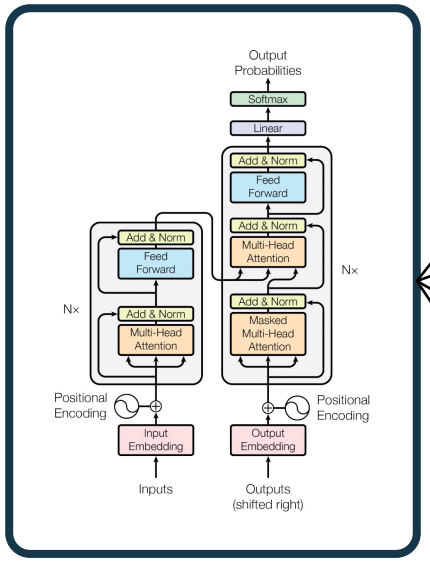
Hallucination!

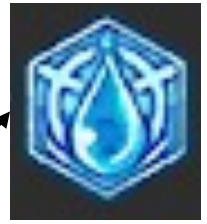
Privacy?!

# We Also Cares! Trustworthy ML

**We = magician**

**ML Models**



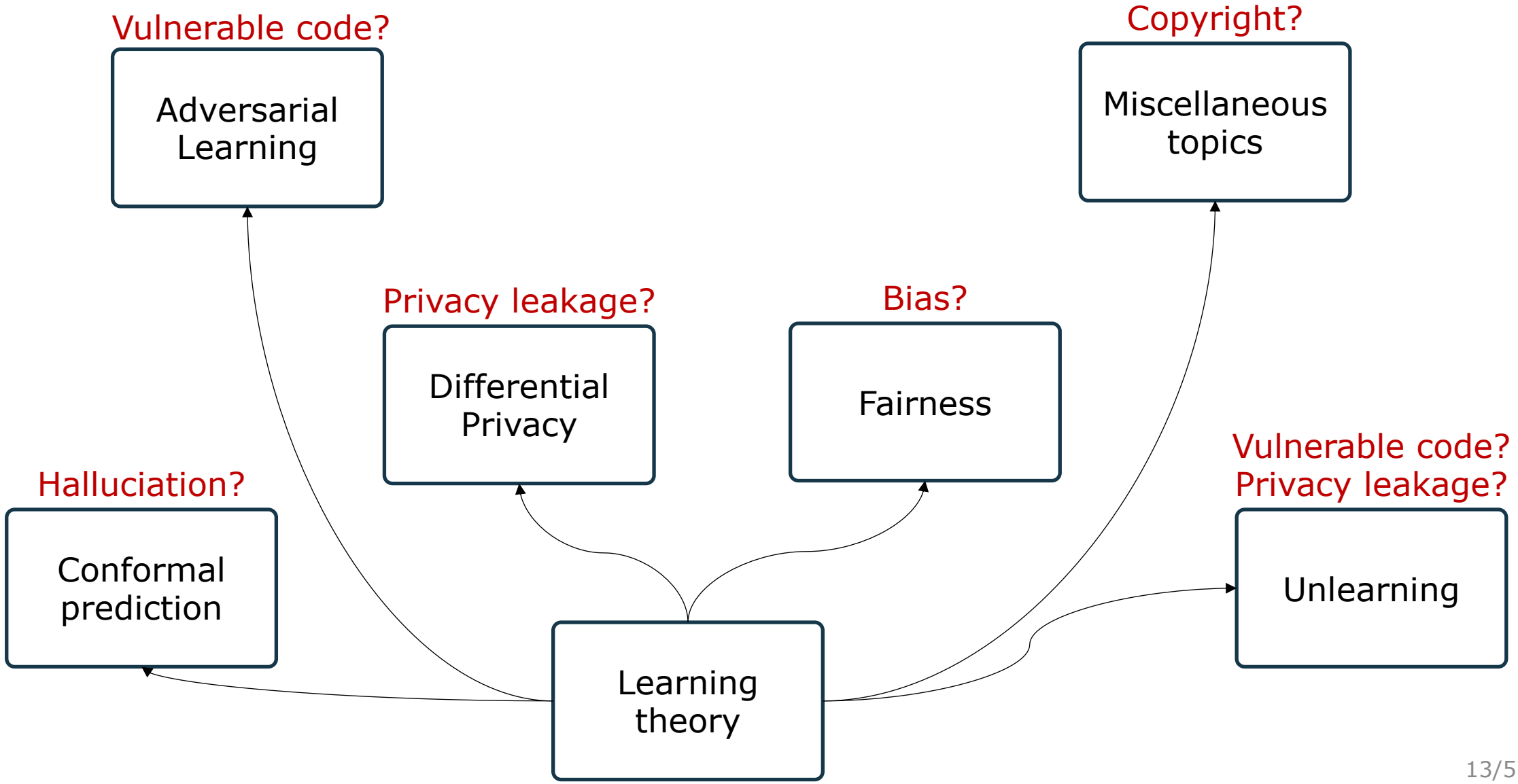
**Self-aware**

**Secure**

**Private**

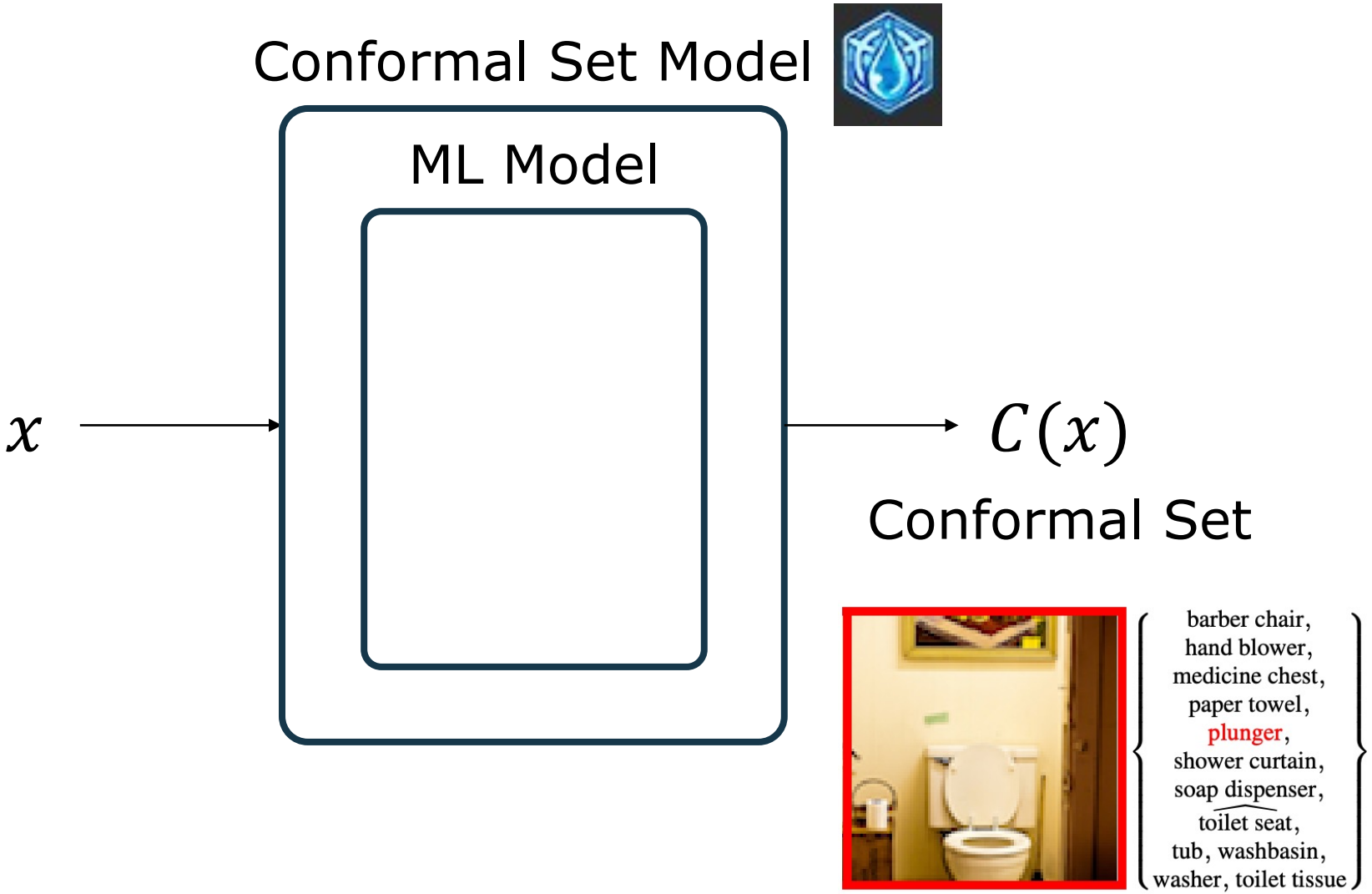
**Fair**

# What We Will Learn (Tentative)



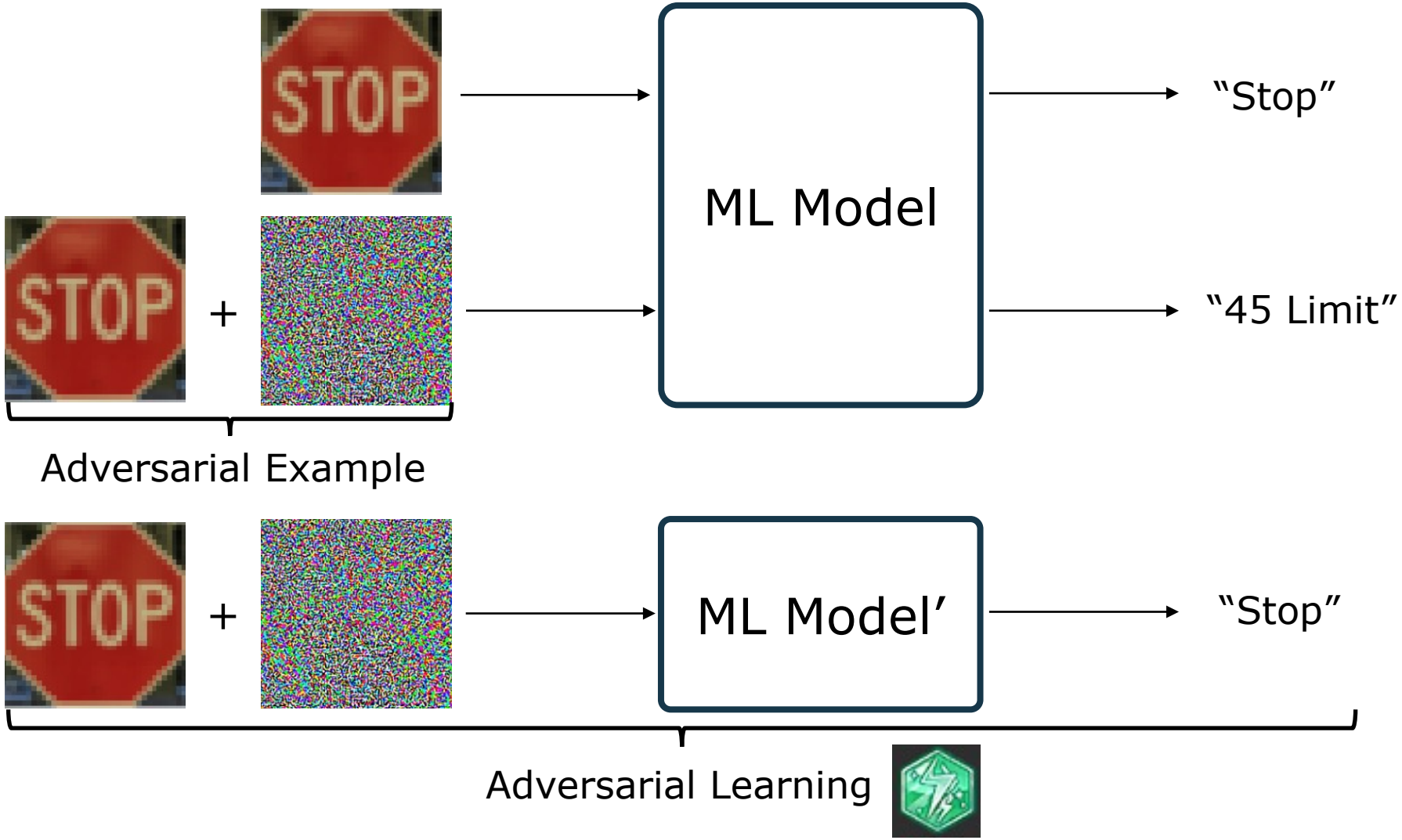
# Conformal Prediction

How to Learn Uncertainty?



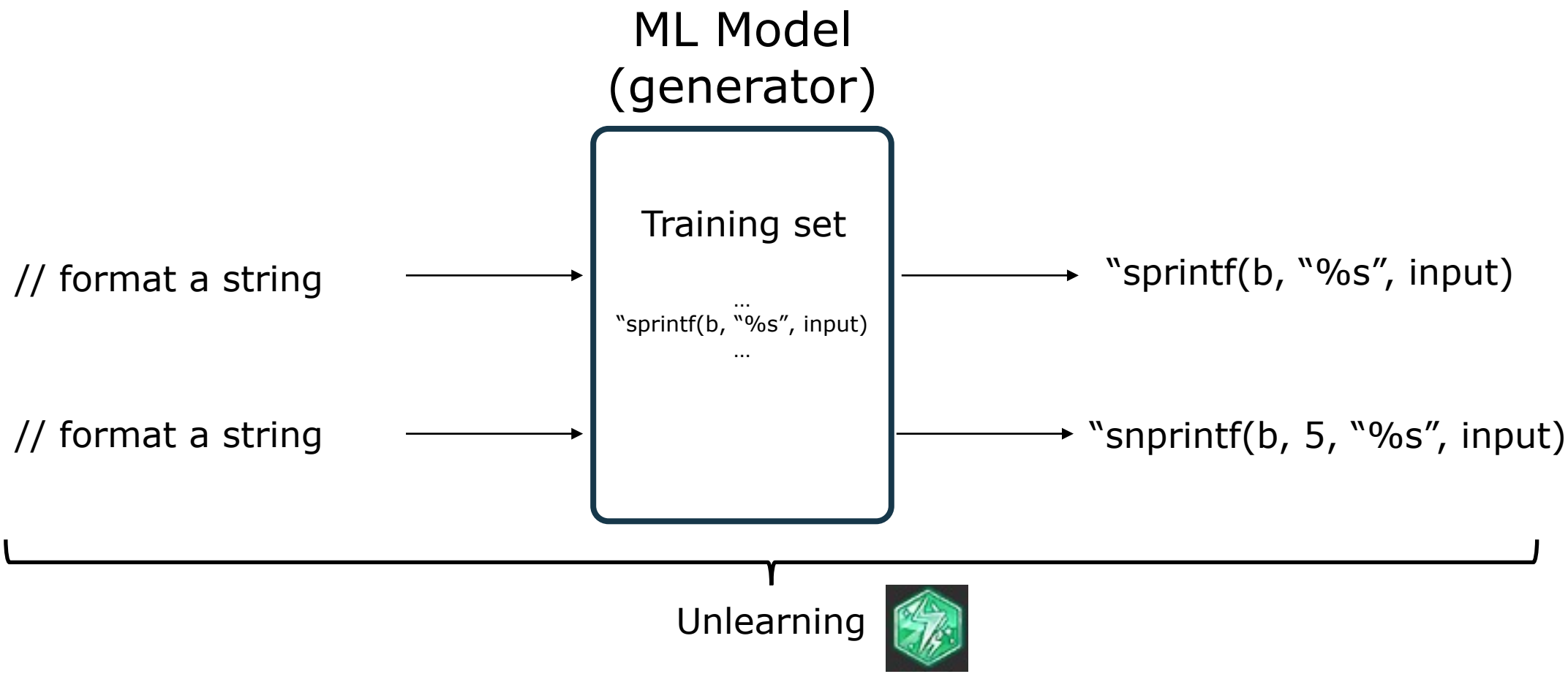
# Adversarial Examples/Learning (=Robustness)

How to learn a model robust to adversarial perturbations?



# Unlearning

How to “relearn” a model to forget learning examples



# Differetial Privacy*

How to learn a model to be "private"?

Sensitive Dataset ("API Key=ABCD") → Learning → ML Model (generator) → generate → "API Key=ABCD"

Sensitive Dataset ("API Key=ABCD") → Differentially private Learning → ML Model (generator) → generate → e.g., "API Key=GDCE"

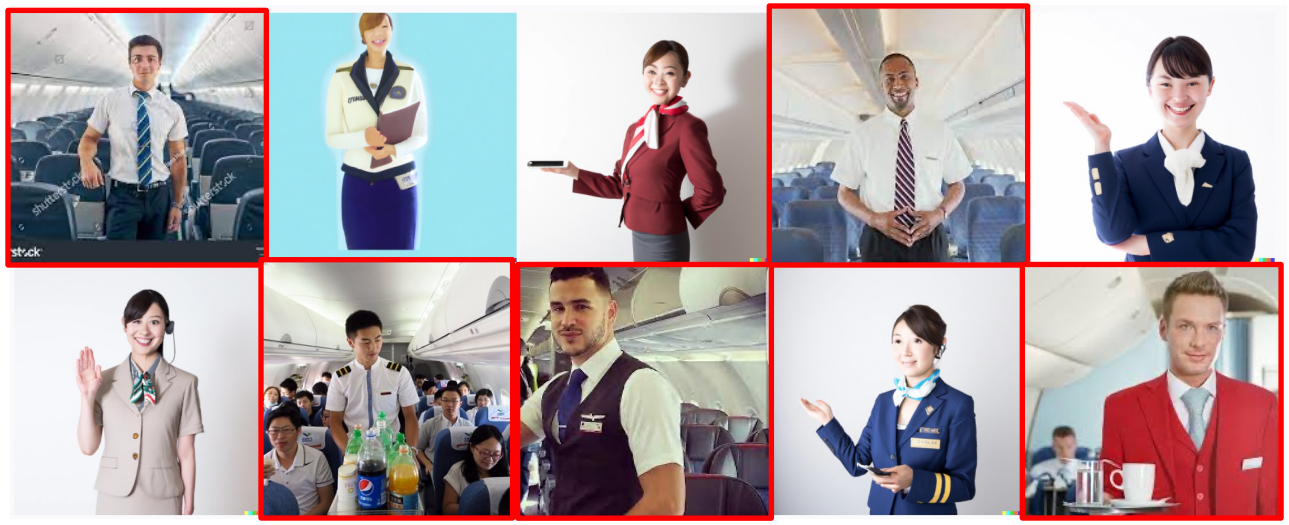
*Differential privacy is more general than learning a model

# Fairness

How to learn a model to be “fair”?

“a flight attendant” → ML Model (text-to-image generator) →

“a flight attendant” → “Fair” ML Model (text-to-image generator) →



https://github.com/openai/dalle-2-preview/blob/eeec5a1843b1d17cb9ed113117a2fcaa9206a564/system-card.md#bias-and-representation

# Miscellaneous Topics on Trustworthy Generative AI

How to avoid copyright issues?

Original art work

ML Model (text-to-image generator)

"Copied" art work

Original art work

+

Cloaked art work

ML Model (text-to-image generator)

"copy-failed" art work

S. Shan et al. "Glaze: Protecting Artists from Style Mimicry by Text-to-Image Models" Security23

# Grading (Tentative)

- Discussion (and attendance): 50%
  - Attentandance
  - Class discussion

- Presentation: 50%
  - Mid presentation
  - Final presentation

- Auditing
  - Feel free to do so